Harsh Tita

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EDUCATION

Arizona State University — Tempe, Arizona, USA
Master of Science, Computer Science (GPA: 4.0/4.0) — Aug 2025 – May 2027
Coursework: Distributed Database Systems, Semantic Web Mining, Knowledge Representation and Reasoning, Data Mining

Amity University — Kolkata, India
Bachelor of Technology, Computer Science and Engineering (GPA: 8.64/10.0) — Jul 2019 – Jun 2023
Coursework: Data structures and Algorithms, Software Engineering, Pattern Recognition, Database Systems, Cloud Computing

TECHNICAL SKILLS

Programming Languages: C/C++, Python, Golang, Java, JavaScript, SQL
Backend & Systems: REST APIs, Distributed Systems, ETL Pipelines, Kafka, Hadoop
AI / Machine Learning: PyTorch, TensorFlow, Scikit-Learn, Hugging Face, LLMs, RAG, Agentic AI
Cloud & Databases: AWS (Bedrock, AgentCore, Lambda, S3, EC2, EMR), PostgreSQL, MongoDB, CockroachDB
Frameworks: Golang Gin, React, Angular
Developer Tools: Docker, Kubernetes, GitLab, Claude Code, Cursor, Kiro

WORK EXPERIENCE

T-Mobile USA, Inc. — Frisco, Texas
-Associate Engineer Intern, AI — May 2026 – Present
Technologies: Python, FastAPI, Cohere, S3 Vector Store, Azure Data Warehouse, Next.js, AWS (Lambda, ECS Fargate, EventBridge)

- Projected to save $14M annually in deferred value by developing an agentic AI delivery prediction system using Cohere embeddings and S3 vector store to surface similar historical Jira capabilities, enabling portfolio managers to make data-driven intake decisions based on past go-live slippages, bugs, and delivery patterns.
- Automated daily vector store updates by engineering serverless pipeline with AWS Lambda and EventBridge Scheduler to process capability data from Azure Data Warehouse, maintaining cost-efficient real-time similarity matching for incoming assessments.
- Architected end-to-end serverless solution with Next.js frontend and AWS ECS Fargate backend using S3 vector store to process historical capability embeddings for portfolio planning decisions.

ASU AI Cloud Innovation Center (powered by Amazon Web Services) — Scottsdale, Arizona
-AI Full-Stack Developer (Student Part-time Intern) — Jan 2026 – May 2026
Technologies: AWS Bedrock, Strands Agents, Python, AWS CDK, Lambda, S3, SES, RAG, IAM

- Reduced AI infrastructure costs by 90% (from $700 to $70/month) and improved response accuracy from 20% to 80% by engineering a custom web crawler using Python and AWS services, replacing generic AWS web crawling solutions with focused data extraction that reduced processing time from 2 hours to 15 minutes.
- Decreased customer inquiry response time from 24 hours to 5 minutes and achieved 95% admin approval rate for AI-generated responses by designing an agentic AI email assistant for Kansas City Planning & Development Department using AWS Bedrock, Strands Agents, and RAG architecture with queue-based processing and multi-step query decomposition.
- Reduced infrastructure deployment time from 8 hours to 15 minutes by automating end-to-end provisioning of 8 AWS services using AWS CDK (Infrastructure as Code), orchestrating Lambda functions, S3 knowledge bases, SES email routing, IAM roles, and Bedrock agent configurations.

ZS Associates — Pune, India
-Software Engineer — Nov 2023 – Jun 2025
Technologies: Python, AWS (EC2, EMR), SQL, Hadoop, Angular, Gen-AI

- Deployed Gen-AI summarization pipeline on AWS EC2 transforming healthcare metrics into insights via Python backend and Angular frontend, increasing user engagement by 60%.
- Engineered automated data validation pipelines using Python and SQL, reducing manual QA effort by 80% across multiple data sources.
- Eliminated weekend manual operations by automating ETL workflows with cron jobs, AWS EMR, and S3-based log monitoring via boto3, replacing monthly laptop-dependent processes with remote status tracking.
- Led automation initiative and scaled impact by onboarding 4 team members, expanding automated workflow coverage across daily, weekly, and monthly pipelines with email notification systems.

PROJECTS

ReEarth: AI-Powered Sustainable Sharing Platform — InnovationHacks 2.0 Hackathon — [Github] — Apr 2026

- Architected full-stack AI sustainability marketplace with 12 AWS Lambda functions using Amazon Bedrock for borrow/buy/skip recommendations, real-time CO tracking, and sustainability scoring across Next.js and DynamoDB backend.
- Deployed serverless infrastructure via AWS CDK integrating Auth0, Stripe payments, ElevenLabs voice search, Amazon Location Service, and S3 storage; collaborated with 2 teammates on end-to-end architecture.

Stance Detection with Large Language Models (LLMs) — [Report] — Nov 2025

- Achieved 80.02% accuracy by fine-tuning LLMs (Phi-3 Mini, Mistral 7B, Llama 3.1 8B) on SemEval-2016 dataset using Parameter-Efficient Fine-Tuning (PEFT) with LoRA and advanced prompt engineering techniques, outperforming baseline approaches.